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STAT 560 PROJECT

ON

**Customer Segmentation And Marketing Strategies For Treadmills:
Analyzing Preferences, Demographics, And Usage Patterns At
Cardio Good Fitness**

INTRODUCTION

The wellness sector is a rapidly changing environment, continually adapting to satisfy the varied demands of health-aware clients. In this competitive arena, staying ahead necessitates a profound comprehension of your customer base. To optimize sales and target the right audience, this project analyzes Cardio Good Fitness's treadmill customer data, aiming to develop detailed buyer personas for each model. This translates to understanding not just who purchases their treadmills, but also why they prefer certain models. By examining customer data, Cardio Good Fitness can obtain valuable insights to improve sales, reach the appropriate audience, and ultimately win over potential treadmill buyers' preferences (and finances). This project explores an examination of customer trends in treadmill purchases at Cardio Good Fitness over the last three months. Our objective is to create in-depth customer profiles for each treadmill model (TM195, TM498, TM798) and use these insights to enhance our marketing strategies and recommendations for sales.

DATA DESCRIPTION

The variables to be examined include:

- **Product purchased (TM195, TM498, or TM798):** This specifies which treadmill model the customer has selected. Each model is tailored to meet distinct user preferences, requirements, and budget constraints.
- **Age (in years):** This specifies the customer's age, which can impact fitness levels, workout routines, and the preferred features or programs in a treadmill.
- **Education (in years):** This quantifies the number of years of formal education completed by the customer. Education level may be associated with income levels and awareness of health-related matters.
- **Marital status (single or partnered):** This denotes whether the customer is single or in a relationship. Relationship status can affect fitness routines and choices regarding purchases.
- **Gender:** This denotes whether the customer identifies as male or female. Gender can occasionally impact fitness objectives, equipment preferences, and exercise routines.
- **Annual household income (\$):** This reflects the total income earned by all members of the customer's household per year. Income can affect affordability and the readiness to invest in fitness equipment.
- **The average number of times the customer plans to use the treadmill each week:** This represents the frequency of treadmill usage per week, indicating the customer's dedication to fitness and their perceived value of owning a treadmill.
- **Average number of miles the customer expects to walk/run each week:** This quantifies the projected weekly distance the customer plans to cover on the treadmill, reflecting their fitness goals and exercise intensity.
- **Self-rated fitness on a 1-to-5 scales (1 = poor shape, 5 = excellent shape) (Fitness):** This is a subjective assessment provided by the customer to evaluate their current fitness level. It offers insight into how they perceive their health and fitness, influencing their exercise routines and preferences for equipment.

EXPLORATORY DATA ANALYSIS

The dataset consists of 180 rows and 9 columns, presenting a detailed overview of the collected information. Each of the 180 rows signifies a distinct observation or entry, documenting specific instances relevant to the study. The 9 columns in the dataset encompass a range of attributes and details, including demographic data and key metrics pertinent to the analysis. These columns cover aspects such as age, gender, education level, marital status, and other pertinent variables. The arrangement of this dataset enables a thorough examination of the data, supporting both descriptive and inferential analysis to derive valuable insights.

I conducted a thorough examination of the dataset's structure, revealing no missing data. Categorical variables, including Marital Status, Gender, and Product, were differentiated from numerical variables such as Age, Education, Usage, Fitness, Income, and Miles. This distinction proved crucial for selecting appropriate analytical approaches. Categorical data necessitated frequency distributions and cross-tabulations to uncover patterns, while numerical data required the calculation of central tendency and dispersion measures. This foundational step was essential for subsequent in-depth exploration and analysis of the dataset.

In the course of exploratory data analysis, I examined the following variables:

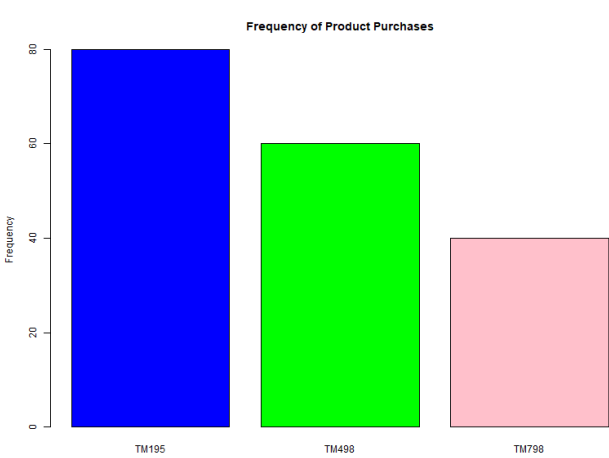


Figure 1.Histogram of Product Purchase

I analyzed the categorical variable of treadmill products (TM195, TM498, and TM798) to understand their frequency distribution. I counted the occurrences of each product i.e (80 (44.44%), 60 (33.33%), and 40 (22.22%) of TM195, TM498, and TM798 were bought respectively) and visualized the distribution using a bar plot as shown in Figure 1. Clearly we can see, there were 3 types of unique models in the store i.e TM195, TM498, and TM798. The most bought treadmill was TM195. The average age of customers purchasing the TM195 model is

28.55 years old.

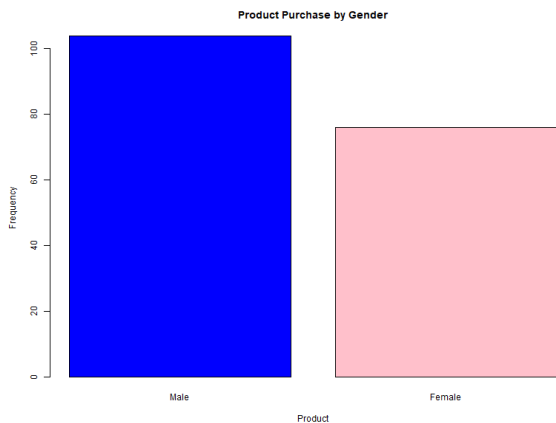


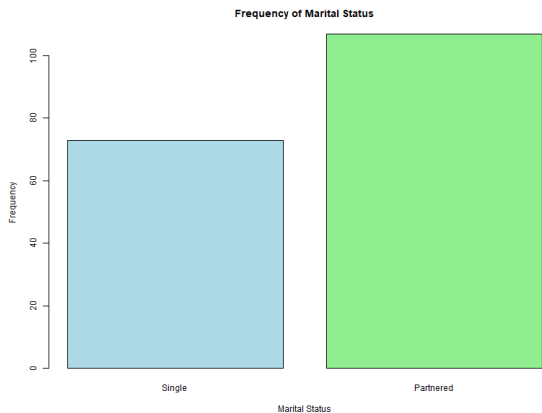
Figure 2. Histogram of Gender

In Figure 2, Male customers dominated treadmill purchases, accounting for 104 out of the 180 sales. This gender disparity highlights a potential area for further investigation to understand factors influencing product preference.

The educational backgrounds of customers displayed notable variability within the dataset. The range of educational attainment extended from 12 to 21 years, reflecting a diverse spectrum of academic qualifications among customers. The median education level was found to be 16 years, indicating that a substantial portion of the customer base had achieved a higher secondary education or some level of tertiary education. These insights into educational demographics offer a clear understanding of the academic qualifications prevalent among the customers.

Based on the summary, the average age of treadmill purchasers is 28.79 years, with an average education level of 15.57 years. Additionally, the average household income of these customers is \$53,720, reflecting a moderate to high earning capacity within the customer base. These demographic characteristics provide a foundational understanding of the target market and can be leveraged to refine marketing strategies and product offerings.

As illustrated in Figure 3, partnered individuals represent the majority of treadmill purchasers, with 107 partnered customers compared to 73 single customers. This observation suggests couples are more inclined to invest in fitness equipment than single individuals. This insight can be utilized marketing strategies and product offerings to better address the preferences and needs of partnered consumers.



that
home
to tailor

Figure 3. Histogram of Marital Status

Treadmill customers report a moderate level of perceived fitness, with an average self-rated fitness score of 3.31 on a 1-to-5 scale. On average, customers expect to use the treadmill approximately 3.46 times per week and cover 103.2 miles weekly. These metrics provide valuable insights into customer exercise habits and expectations, which can be instrumental in guiding product development and tailoring marketing strategies.

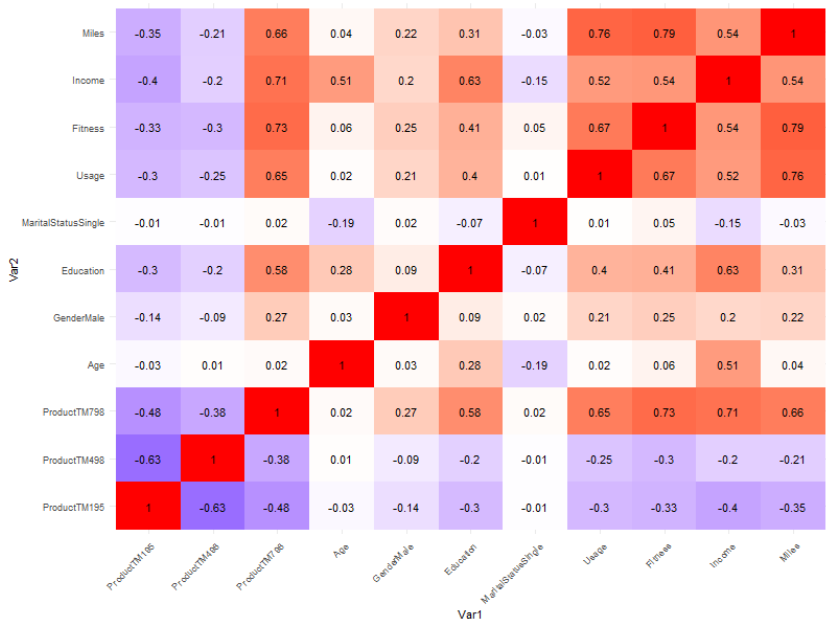


Figure 4. Correlation Matrix

There is a strong positive correlation between several key variables: fitness and usage, fitness and income, miles and usage, product TM798 and fitness, product TM798 and usage, and product TM498 and income. These correlations suggest that customers with higher fitness levels tend to use the treadmill more frequently, have higher incomes, and are more likely to purchase higher-end models like the TM798 and TM498. This information is crucial for understanding customer behavior and preferences,

aiding in targeted marketing and product positioning strategies. There is a strong negative correlation between the TM195 model and several key variables such as income, fitness, and usage. This indicates that customers who purchase the TM195 model tend to have lower incomes, lower fitness levels, and use the treadmill less frequently. Understanding these patterns can help tailor marketing efforts and product offerings to better meet the needs of this customer segment.

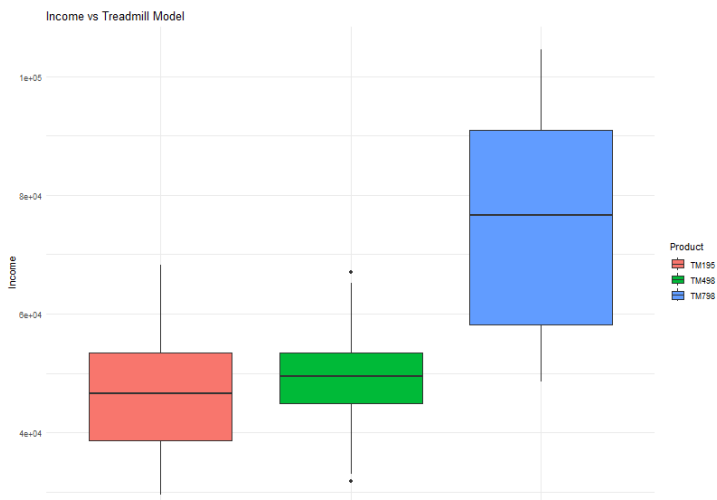


Figure 5. boxplot of income vs treadmills

Figure 5 illustrates the income distribution across different treadmill models. As previously discussed, customers who purchased the TM798 model tend to have significantly higher incomes compared to those who bought the TM195 and TM498 models. Conversely, customers who opted for the TM195 model tend to have lower incomes. This visual representation reinforces the

relationship between income level and product choice, clearly distinguishing among the three customer segments.

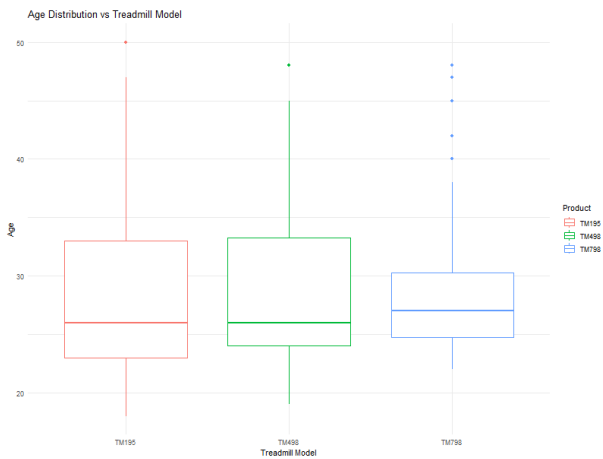


Figure 6. Boxplot of Age vs Treadmills

There are some outliers, particularly in the higher age range, for all three models, indicating that a few older customers are purchasing each type of treadmill. Additionally, the distribution of ages within each model group seems somewhat skewed to the right, with a larger proportion of younger customers. This visual comparison highlights the central tendency, variability, and potential outliers in the age data for each model, allowing for a clear understanding of the demographic differences among customers who prefer each treadmill model. These insights can inform targeted marketing strategies and product development decisions.

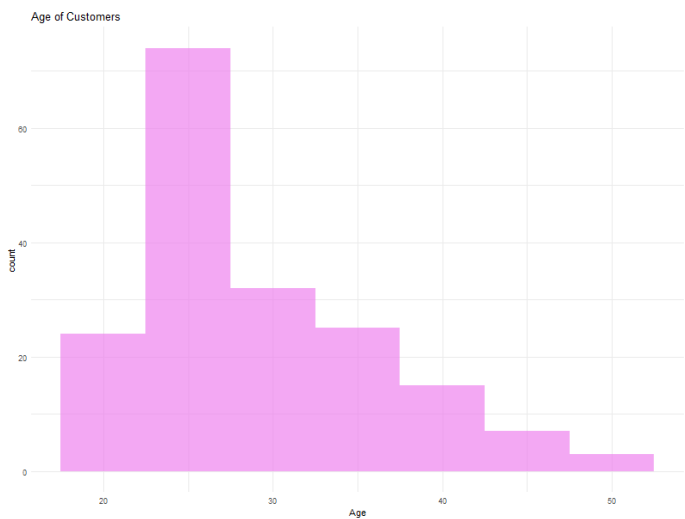


Figure 7 Histogram of age of customers

The provided box plot in Figure 6 visually compares the age distribution of customers who purchased different treadmill models (TM195, TM498, and TM798). The age range of customers purchasing all three models is relatively similar, from approximately 20 to 50 years. The median age for customers purchasing the TM195 model appears slightly lower than that of the other two models, suggesting a younger demographic for TM195 buyers.

Figure 7 presents a histogram of customer age distribution, highlighting that the majority of customers are in the 25-30 age range. There is a noticeable drop in the number of customers as age increases beyond 30, with fewer customers in the 20-25 age range and progressively fewer from 30 to 50 years. This indicates that the primary customer base consists mainly of individuals in their mid-20s to early 30s. Marketing and product development should focus on this demographic, while also exploring

strategies to attract older age groups.

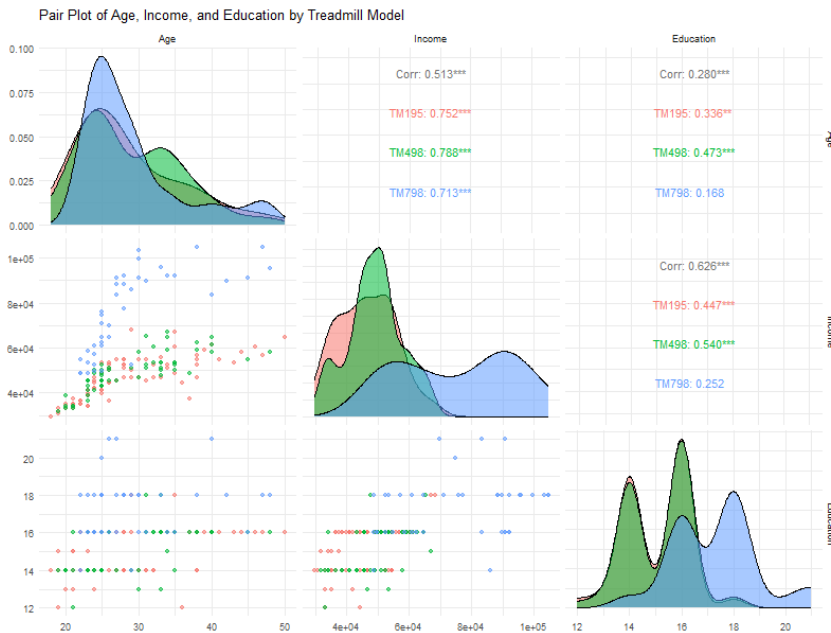


Figure 8 pair plot of age, income and education by product

Figure 8 displays a pair plot illustrating the relationships between age, income, and education for customers, differentiated by treadmill model (TM195 in red, TM498 in green, and TM798 in blue). The diagonal panels show the distribution (density plots) of each variable, revealing that TM498 customers generally have higher income and education levels. The upper off-diagonal panels display the correlation coefficients between the variables, indicating significant

positive correlations, particularly between income and age (Corr: 0.513) and between income and education (Corr: 0.626). The lower off-diagonal panels provide scatter plots of the variable pairs, demonstrating positive trends and clustering by treadmill model, with TM498 customers forming distinct groups with higher values in income and education. This pair plot helps to visualize the distribution and interrelationships of these variables, highlighting distinct demographic characteristics associated with each treadmill model.

Figure 9 shows a box plot comparing fitness scores across different treadmill models. TM798 exhibits a significantly higher and more variable range of fitness scores compared to TM195 and TM498, which both have lower and nearly identical fitness scores. This suggests that users of TM798 generally rate their fitness much higher than those using the other two models. The greater variability in fitness scores for TM798 also indicates that this model appeals to a broader spectrum of fitness levels, potentially attracting users with higher fitness aspirations.

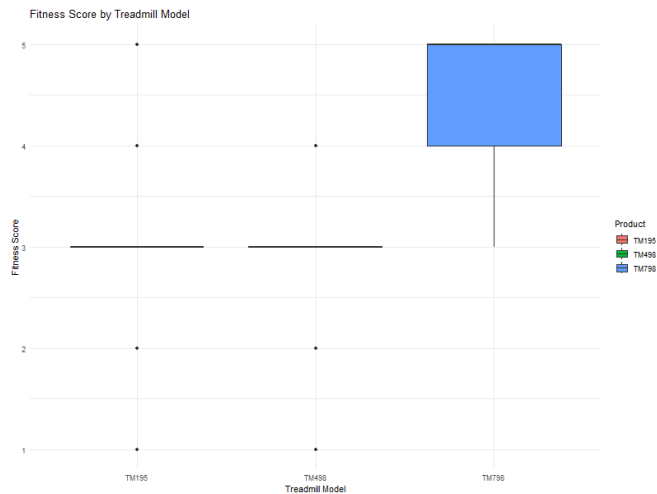


Figure 9 boxplot of fitness score vs product

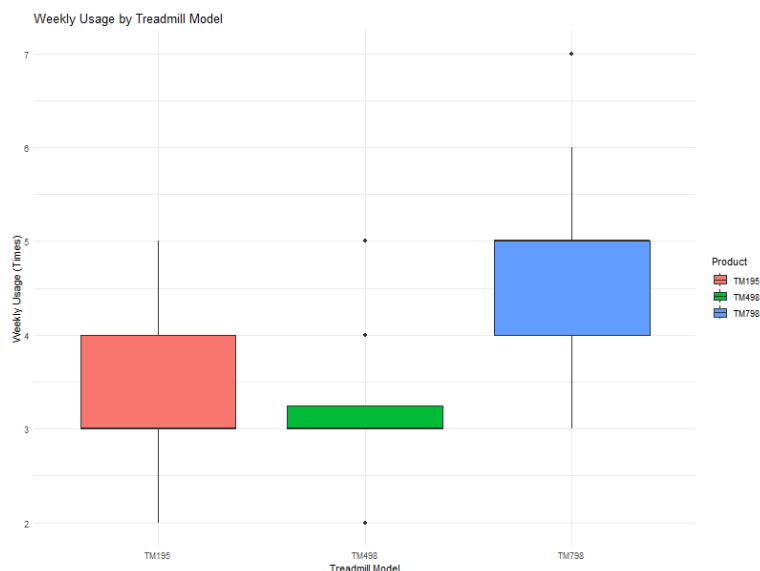


Figure 10 boxplot of weekly usage vs product

Figure 10 shows a box plot comparing weekly treadmill usage across different models. TM195 exhibits moderate weekly usage with some variability, TM498 has the lowest and most consistent usage, and TM798 demonstrates the highest and most variable usage. This indicates that users of TM798 tend to use their treadmills more frequently and with greater variation compared to users of TM195 and TM498.

CONCLUSION AND RECOMMENDATIONS

Based on the analysis, TM195 emerges as the most popular treadmill model, representing 44.44% of sales. This model should be marketed as a reliable and budget-friendly option, highlighting its affordability and essential features to attract a broad range of customers. TM195 is associated with lower income levels, indicating that it is a preferred choice for cost-conscious consumers seeking value for money.

In contrast, TM798 stands out as the model attracting higher-income customers. It also shows significantly higher and more variable fitness scores, suggesting that it appeals to those looking for advanced features and a premium fitness experience. Marketing TM798 as a high-end, luxurious treadmill with superior capabilities will cater to professionals and higher-income individuals willing to invest more in their fitness equipment.

TM498, associated with higher income and education levels compared to TM195, represents a mid-range option. It should be promoted as a treadmill that balances affordability and enhanced features, appealing to customers seeking quality without the highest price tag.

Furthermore, TM798 users exhibit the highest and most variable treadmill usage, indicating strong engagement and varied fitness goals. Emphasizing the versatility and advanced features of TM798 can attract users with diverse fitness needs and higher aspirations. By aligning marketing strategies with these observations, Cardio Good Fitness can effectively target different customer segments, improve product positioning, and enhance overall sales and satisfaction.

APPENDIX

R CODES

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2     3.5.1      v tibble    3.2.1
## v lubridate  1.9.3      v tidyr     1.3.1
## v purrr       1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)
library(readr)
library(dplyr)
library(lattice)
#Loading data
data=read.csv("CardioGoodFitness.csv")
#View(data)
glimpse(data)
```

```
## Rows: 180
## Columns: 9
## $ Product      <chr> "TM195", "TM195", "TM195", "TM195", "TM195", "TM195", "T~
## $ Age          <int> 18, 19, 19, 19, 20, 20, 21, 21, 21, 21, 22, 22, 22, 22, ~
## $ Gender       <chr> "Male", "Male", "Female", "Male", "Male", "Female", "Fem~
## $ Education    <int> 14, 15, 14, 12, 13, 14, 14, 13, 15, 15, 14, 14, 16, 14, ~
## $ MaritalStatus <chr> "Single", "Single", "Partnered", "Single", "Partnered", ~
## $ Usage        <int> 3, 2, 4, 3, 4, 3, 3, 3, 5, 2, 3, 3, 4, 3, 3, 3, 2, 4, 4,~
## $ Fitness      <int> 4, 3, 3, 3, 2, 3, 3, 3, 4, 3, 3, 2, 3, 3, 1, 3, 3, 3, 3,~
## $ Income       <int> 29562, 31836, 30699, 32973, 35247, 32973, 35247, 32973, ~
## $ Miles        <int> 112, 75, 66, 85, 47, 66, 75, 85, 141, 85, 85, 66, 75, 75~
```

```
str(data)
```

```
## 'data.frame':   180 obs. of  9 variables:
## $ Product      : chr  "TM195" "TM195" "TM195" "TM195" ...
## $ Age          : int   18 19 19 19 20 20 21 21 21 21 ...
## $ Gender       : chr   "Male" "Male" "Female" "Male" ...
## $ Education    : int   14 15 14 12 13 14 14 13 15 15 ...
## $ MaritalStatus: chr   "Single" "Single" "Partnered" "Single" ...
## $ Usage        : int    3 2 4 3 4 3 3 3 5 2 ...
## $ Fitness      : int    4 3 3 3 2 3 3 3 4 3 ...
## $ Income       : int  29562 31836 30699 32973 35247 32973 35247 32973 35247 37521 ...
## $ Miles        : int   112 75 66 85 47 66 75 85 141 85 ...
```

```
mean(is.na(data))
```

```
## [1] 0
```

```
#converting data to the format thats numerics
```

```
data <- data %>%
```

```
  mutate(
    Product = factor(Product, levels = c("TM195", "TM498", "TM798")),
    Gender = factor(Gender, levels = c("Male", "Female")),
    MaritalStatus = factor(MaritalStatus, levels = c("Single", "Partnered"))
  )
```

```
# Check the structure of the data to confirm changes
```

```
str(data)
```

```
## 'data.frame': 180 obs. of 9 variables:
## $ Product : Factor w/ 3 levels "TM195","TM498",...: 1 1 1 1 1 1 1 1 1 ...
## $ Age : int 18 19 19 19 20 20 21 21 21 ...
## $ Gender : Factor w/ 2 levels "Male","Female": 1 1 2 1 1 2 2 1 1 2 ...
## $ Education : int 14 15 14 12 13 14 14 13 15 15 ...
## $ MaritalStatus: Factor w/ 2 levels "Single","Partnered": 1 1 2 1 2 2 2 1 1 2 ...
## $ Usage : int 3 2 4 3 4 3 3 3 5 2 ...
## $ Fitness : int 4 3 3 3 2 3 3 3 4 3 ...
## $ Income : int 29562 31836 30699 32973 35247 32973 35247 32973 35247 37521 ...
## $ Miles : int 112 75 66 85 47 66 75 85 141 85 ...
```

```
str(data)
```

```
## 'data.frame': 180 obs. of 9 variables:
## $ Product : Factor w/ 3 levels "TM195","TM498",...: 1 1 1 1 1 1 1 1 1 ...
## $ Age : int 18 19 19 19 20 20 21 21 21 ...
## $ Gender : Factor w/ 2 levels "Male","Female": 1 1 2 1 1 2 2 1 1 2 ...
## $ Education : int 14 15 14 12 13 14 14 13 15 15 ...
## $ MaritalStatus: Factor w/ 2 levels "Single","Partnered": 1 1 2 1 2 2 2 1 1 2 ...
## $ Usage : int 3 2 4 3 4 3 3 3 5 2 ...
## $ Fitness : int 4 3 3 3 2 3 3 3 4 3 ...
## $ Income : int 29562 31836 30699 32973 35247 32973 35247 32973 35247 37521 ...
## $ Miles : int 112 75 66 85 47 66 75 85 141 85 ...
```

```
#View(data)
```

```
summary(data)
```

```
## Product Age Gender Education MaritalStatus
## TM195:80 Min. :18.00 Male :104 Min. :12.00 Single : 73
## TM498:60 1st Qu.:24.00 Female: 76 1st Qu.:14.00 Partnered:107
## TM798:40 Median :26.00 Median :16.00
## Mean :28.79 Mean :15.57
## 3rd Qu.:33.00 3rd Qu.:16.00
## Max. :50.00 Max. :21.00
## Usage Fitness Income Miles
## Min. :2.000 Min. :1.000 Min. : 29562 Min. : 21.0
```

```
## 1st Qu.:3.000 1st Qu.:3.000 1st Qu.: 44059 1st Qu.: 66.0
## Median :3.000 Median :3.000 Median : 50597 Median : 94.0
## Mean :3.456 Mean :3.311 Mean : 53720 Mean :103.2
## 3rd Qu.:4.000 3rd Qu.:4.000 3rd Qu.: 58668 3rd Qu.:114.8
## Max. :7.000 Max. :5.000 Max. :104581 Max. :360.0
```

```
dim(data)
```

```
## [1] 180 9
```

```
#Determining the frequency distribution for categorical variables.
```

```
#For Product
```

```
frequency_table_Product <- table(data$Product)
```

```
print(frequency_table_Product)
```

```
##
```

```
## TM195 TM498 TM798
```

```
## 80 60 40
```

```
total_count <- sum(frequency_table_Product)
```

```
percentage <- prop.table(frequency_table_Product) * 100
```

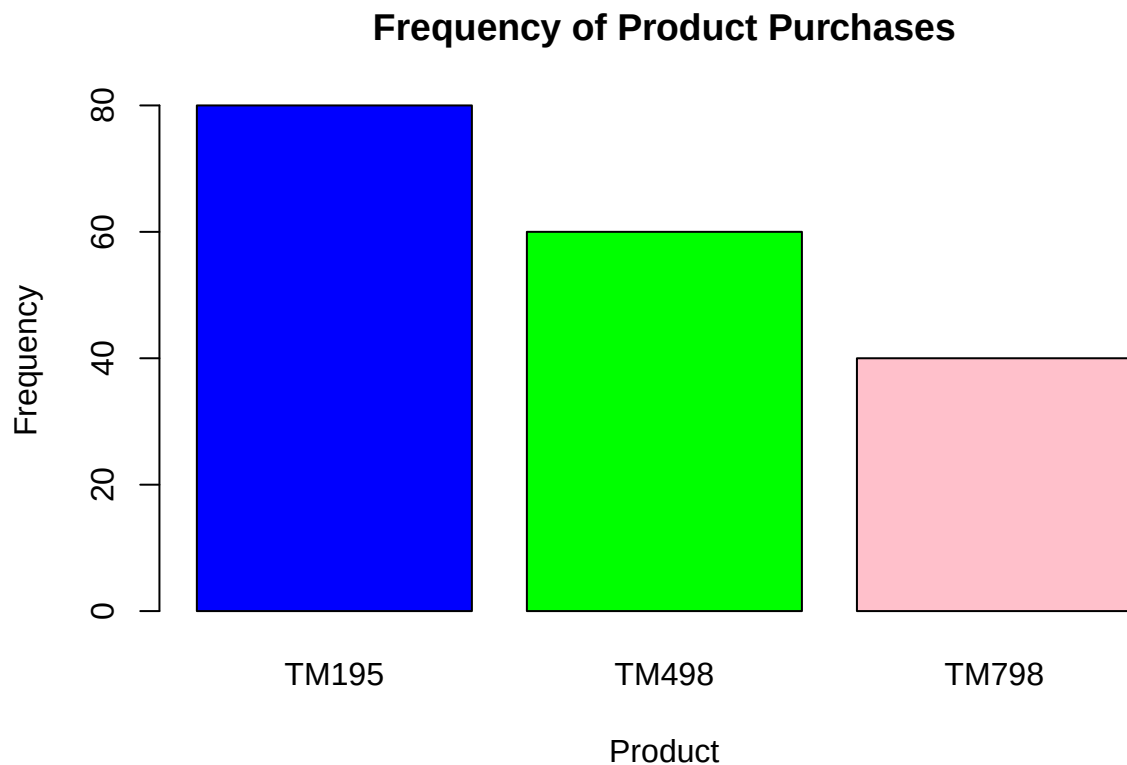
```
print(percentage)
```

```
##
```

```
## TM195 TM498 TM798
```

```
## 44.44444 33.33333 22.22222
```

```
barplot(frequency_table_Product, col = c("blue", "green", "pink"), main = "Frequency of Product Purchases")
```



```
#models
unique_models <- unique(data$Product)
number_of_models <- length(unique_models)
```

```
#For Gender
frequency_table_Gender <- table(data$Gender)
print(frequency_table_Gender)
```

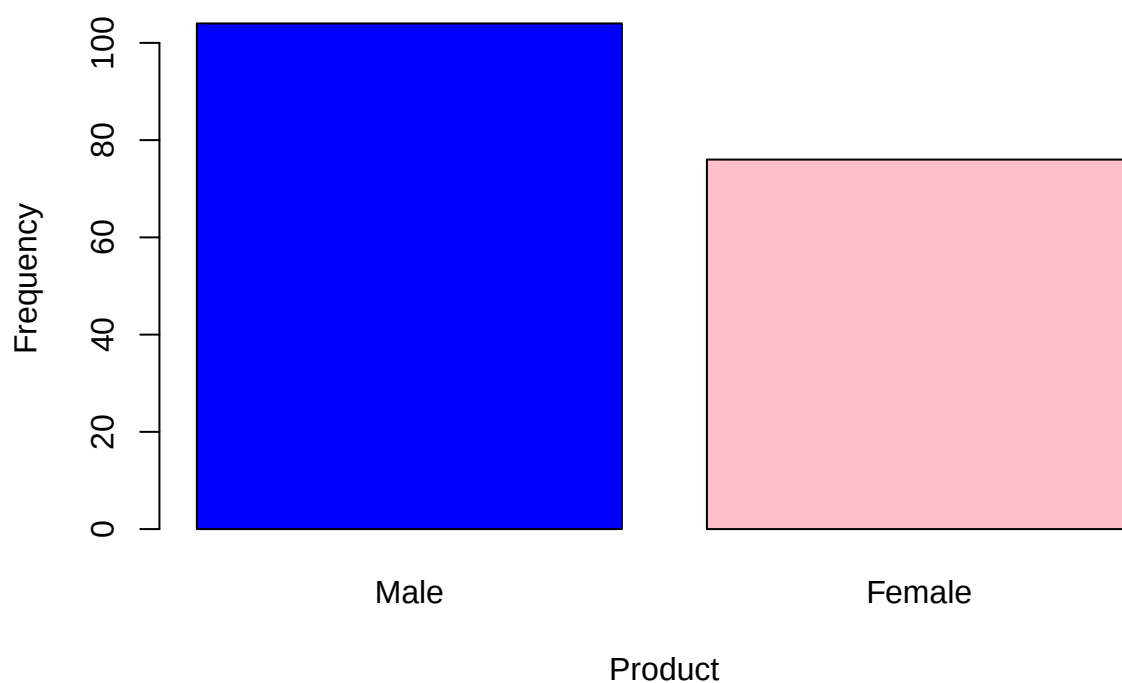
```
##
##   Male Female
##   104     76
```

```
total_count <- sum(frequency_table_Gender)
percentage <- prop.table(frequency_table_Gender) * 100
print(percentage)
```

```
##
##      Male   Female
## 57.77778 42.22222
```

```
barplot(frequency_table_Gender, col = c("blue", "pink"), main = "Product Purchase by Gender", xlab = "Product")
```

Product Purchase by Gender



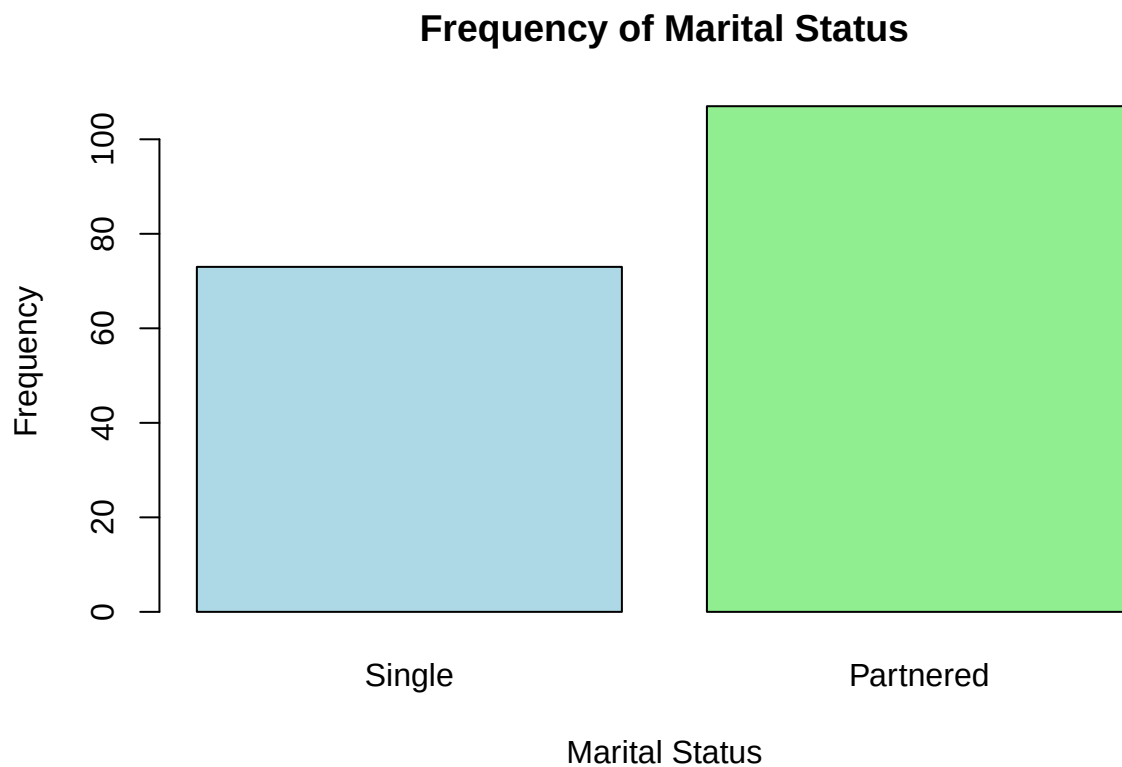
```
#For Marital status
frequency_table_MaritalStatus <- table(data$MaritalStatus)
print(frequency_table_MaritalStatus)
```

```
##
##   Single Partnered
##      73      107
```

```
total_count <- sum(frequency_table_MaritalStatus)
percentage <- prop.table(frequency_table_MaritalStatus) * 100
print(percentage)
```

```
##
##   Single Partnered
## 40.55556 59.44444
```

```
barplot(frequency_table_MaritalStatus,col = c("lightblue", "lightgreen"), main = "Frequency of Marital Status")
```



```
average_age_TM195 <- data %>%
  filter(Product == "TM195") %>%
  summarise(Average_Age = mean(Age, na.rm = TRUE))
# Print the result
print(average_age_TM195)
```

```
## Average_Age
## 1 28.55
```

```
#range of customers
education_range <- range(data$Education, na.rm = TRUE)
education_median <- median(data$Education, na.rm = TRUE)
print(education_range)
```

```
## [1] 12 21
```

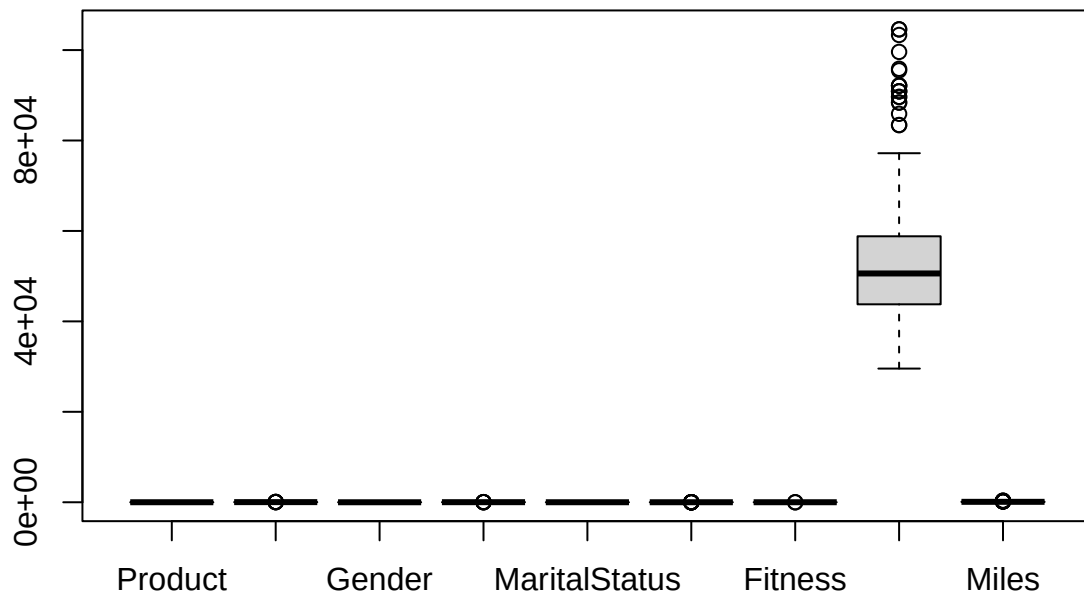
```
print(education_median)
```

```
## [1] 16
```

```
#hist(data$Product, main="Distribution of Product", xlab="product")
```

```
# boxplot for entire dataset after factoring
```

```
boxplot(data, notch = F)
```



```
average_age_TM195 <- mean(data %>% filter(Product == "TM195") %>% select(Age) %>% unlist())
average_age_TM498 <- mean(data %>% filter(Product == "TM498") %>% select(Age) %>% unlist())
average_age_TM798 <- mean(data %>% filter(Product == "TM798") %>% select(Age) %>% unlist())
```

```
average_age_TM195
```

```
## [1] 28.55
```

```
average_age_TM498
```

```
## [1] 28.9
```

```
average_age_TM798
```

```
## [1] 29.1
```

```
library(reshape2)
```

```
##  
## Attaching package: 'reshape2'  
  
## The following object is masked from 'package:tidyr':  
##  
## smiths
```

```
#converting data to the format thats numerics and getting the correlation plot
```

```
df_encoded <- data %>%  
  mutate(across(where(is.factor), as.character)) %>%  
  mutate(across(where(is.character), as.factor)) %>%  
  model.matrix(~ . - 1, data = .) %>%  
  as.data.frame()  
  
df_encoded <- df_encoded[, c(grep("Product", names(df_encoded)), setdiff(seq_along(names(df_encoded)),  
correlation_matrix <- cor(df_encoded)  
print(correlation_matrix)
```

```
##           ProductTM195 ProductTM498 ProductTM798      Age  
## ProductTM195      1.00000000 -0.63245532 -0.47809144 -0.03085833  
## ProductTM498     -0.63245553  1.00000000 -0.37796447  0.01134681  
## ProductTM798     -0.47809144 -0.377964473  1.00000000  0.02401668  
## Age              -0.03085833  0.011346813  0.02401668  1.00000000  
## GenderMale       -0.14084764 -0.087489155  0.26754850  0.02754355  
## Education        -0.29659116 -0.199761285  0.58100224  0.28049567  
## MaritalStatusSingle -0.01012026 -0.008000768  0.02116804 -0.19215177  
## Usage            -0.30431238 -0.254197798  0.65195607  0.01506447  
## Fitness          -0.32608954 -0.304014902  0.73447205  0.06110454  
## Income           -0.39674377 -0.203871267  0.70536779  0.51341369  
## Miles            -0.35291492 -0.208649907  0.65840076  0.03661757  
##           GenderMale  Education MaritalStatusSingle      Usage  
## ProductTM195     -0.14084764 -0.29659116      -0.010120260 -0.304312379  
## ProductTM498     -0.08748915 -0.19976128      -0.008000768 -0.254197798  
## ProductTM798      0.26754850  0.58100224      0.021168043  0.651956072  
## Age              0.02754355  0.28049567      -0.192151767  0.015064469  
## GenderMale       1.00000000  0.09408875      0.018835837  0.214423926  
## Education        0.09408875  1.00000000      -0.068568554  0.395155219  
## MaritalStatusSingle 0.01883584 -0.06856855      1.000000000  0.007786463  
## Usage            0.21442393  0.39515522      0.007786463  1.000000000  
## Fitness          0.25460893  0.41058079      0.050750711  0.668605568  
## Income           0.20205258  0.62582735      -0.150292900  0.519537232  
## Miles            0.21786876  0.30728428      -0.025638980  0.759130477  
##           Fitness      Income      Miles  
## ProductTM195     -0.32608954 -0.3967438 -0.35291492  
## ProductTM498     -0.30401490 -0.2038713 -0.20864991  
## ProductTM798      0.73447205  0.7053678  0.65840076  
## Age              0.06110454  0.5134137  0.03661757  
## GenderMale       0.25460893  0.2020526  0.21786876  
## Education        0.41058079  0.6258273  0.30728428  
## MaritalStatusSingle 0.05075071 -0.1502929 -0.02563898
```



```
## Usage          0.66860557  0.5195372  0.75913048
## Fitness        1.00000000  0.5350053  0.78570174
## Income          0.53500532  1.0000000  0.54347326
## Miles           0.78570174  0.5434733  1.00000000
```

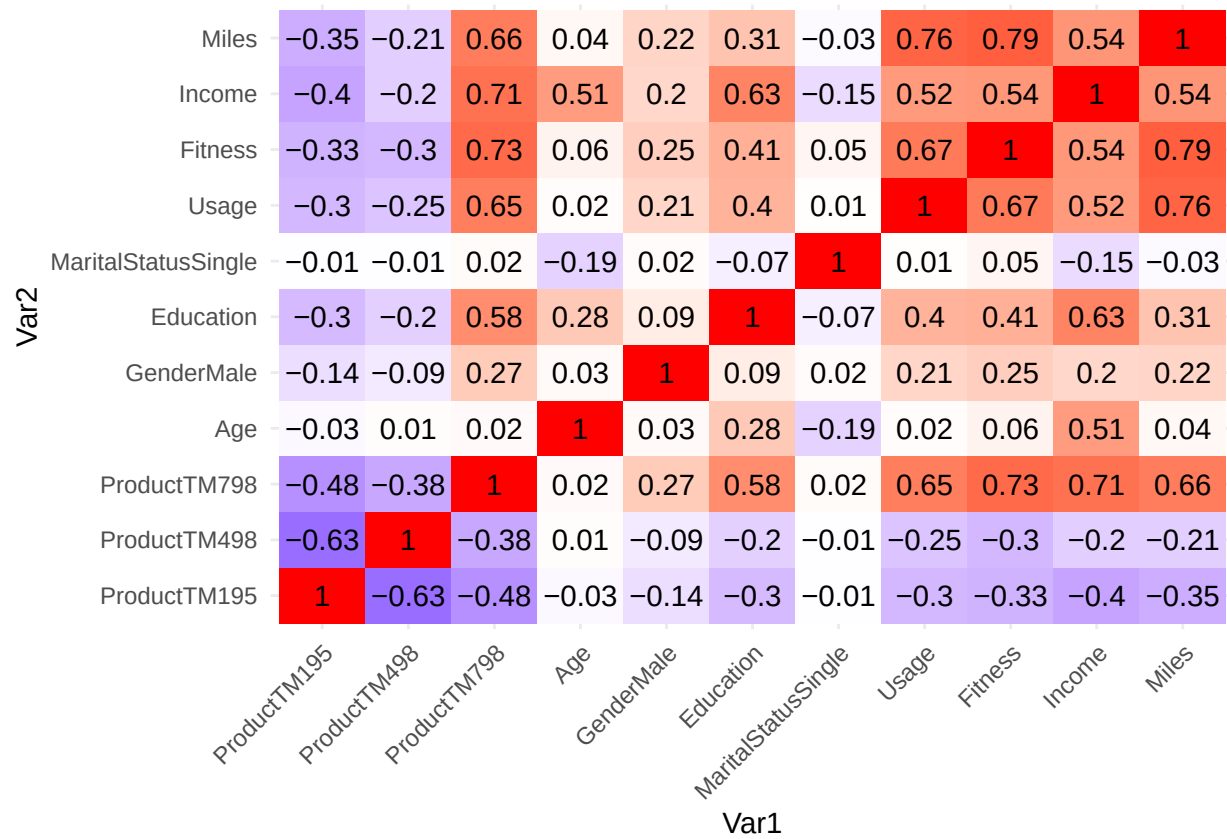
```
melted_cormat <- melt(correlation_matrix)
print(melted_cormat)
```

```
##           Var1           Var2      value
## 1   ProductTM195   ProductTM195  1.000000000
## 2   ProductTM498   ProductTM195 -0.632455532
## 3   ProductTM798   ProductTM195 -0.478091444
## 4           Age    ProductTM195 -0.030858326
## 5   GenderMale    ProductTM195 -0.140847636
## 6   Education      ProductTM195 -0.296591158
## 7   MaritalStatusSingle ProductTM195 -0.010120260
## 8           Usage    ProductTM195 -0.304312379
## 9           Fitness   ProductTM195 -0.326089545
## 10          Income    ProductTM195 -0.396743770
## 11          Miles    ProductTM195 -0.352914916
## 12   ProductTM195   ProductTM498 -0.632455532
## 13   ProductTM498   ProductTM498  1.000000000
## 14   ProductTM798   ProductTM498 -0.377964473
## 15          Age    ProductTM498  0.011346813
## 16   GenderMale    ProductTM498 -0.087489155
## 17   Education      ProductTM498 -0.199761285
## 18   MaritalStatusSingle ProductTM498 -0.008000768
## 19          Usage    ProductTM498 -0.254197798
## 20          Fitness   ProductTM498 -0.304014902
## 21          Income    ProductTM498 -0.203871267
## 22          Miles    ProductTM498 -0.208649907
## 23   ProductTM195   ProductTM798 -0.478091444
## 24   ProductTM498   ProductTM798 -0.377964473
## 25   ProductTM798   ProductTM798  1.000000000
## 26          Age    ProductTM798  0.024016677
## 27   GenderMale    ProductTM798  0.267548502
## 28   Education      ProductTM798  0.581002243
## 29   MaritalStatusSingle ProductTM798  0.021168043
## 30          Usage    ProductTM798  0.651956072
## 31          Fitness   ProductTM798  0.734472050
## 32          Income    ProductTM798  0.705367792
## 33          Miles    ProductTM798  0.658400761
## 34   ProductTM195           Age -0.030858326
## 35   ProductTM498           Age  0.011346813
## 36   ProductTM798           Age  0.024016677
## 37           Age           Age  1.000000000
## 38   GenderMale           Age  0.027543554
## 39   Education           Age  0.280495667
## 40   MaritalStatusSingle Age -0.192151767
## 41          Usage           Age  0.015064469
## 42          Fitness           Age  0.061104542
## 43          Income           Age  0.513413685
## 44          Miles           Age  0.036617569
## 45   ProductTM195   GenderMale -0.140847636
```

## 46	ProductTM498	GenderMale	-0.087489155
## 47	ProductTM798	GenderMale	0.267548502
## 48	Age	GenderMale	0.027543554
## 49	GenderMale	GenderMale	1.000000000
## 50	Education	GenderMale	0.094088747
## 51	MaritalStatusSingle	GenderMale	0.018835837
## 52	Usage	GenderMale	0.214423926
## 53	Fitness	GenderMale	0.254608931
## 54	Income	GenderMale	0.202052581
## 55	Miles	GenderMale	0.217868763
## 56	ProductTM195	Education	-0.296591158
## 57	ProductTM498	Education	-0.199761285
## 58	ProductTM798	Education	0.581002243
## 59	Age	Education	0.280495667
## 60	GenderMale	Education	0.094088747
## 61	Education	Education	1.000000000
## 62	MaritalStatusSingle	Education	-0.068568554
## 63	Usage	Education	0.395155219
## 64	Fitness	Education	0.410580790
## 65	Income	Education	0.625827346
## 66	Miles	Education	0.307284282
## 67	ProductTM195	MaritalStatusSingle	-0.010120260
## 68	ProductTM498	MaritalStatusSingle	-0.008000768
## 69	ProductTM798	MaritalStatusSingle	0.021168043
## 70	Age	MaritalStatusSingle	-0.192151767
## 71	GenderMale	MaritalStatusSingle	0.018835837
## 72	Education	MaritalStatusSingle	-0.068568554
## 73	MaritalStatusSingle	MaritalStatusSingle	1.000000000
## 74	Usage	MaritalStatusSingle	0.007786463
## 75	Fitness	MaritalStatusSingle	0.050750711
## 76	Income	MaritalStatusSingle	-0.150292900
## 77	Miles	MaritalStatusSingle	-0.025638980
## 78	ProductTM195	Usage	-0.304312379
## 79	ProductTM498	Usage	-0.254197798
## 80	ProductTM798	Usage	0.651956072
## 81	Age	Usage	0.015064469
## 82	GenderMale	Usage	0.214423926
## 83	Education	Usage	0.395155219
## 84	MaritalStatusSingle	Usage	0.007786463
## 85	Usage	Usage	1.000000000
## 86	Fitness	Usage	0.668605568
## 87	Income	Usage	0.519537232
## 88	Miles	Usage	0.759130477
## 89	ProductTM195	Fitness	-0.326089545
## 90	ProductTM498	Fitness	-0.304014902
## 91	ProductTM798	Fitness	0.734472050
## 92	Age	Fitness	0.061104542
## 93	GenderMale	Fitness	0.254608931
## 94	Education	Fitness	0.410580790
## 95	MaritalStatusSingle	Fitness	0.050750711
## 96	Usage	Fitness	0.668605568
## 97	Fitness	Fitness	1.000000000
## 98	Income	Fitness	0.535005317
## 99	Miles	Fitness	0.785701743

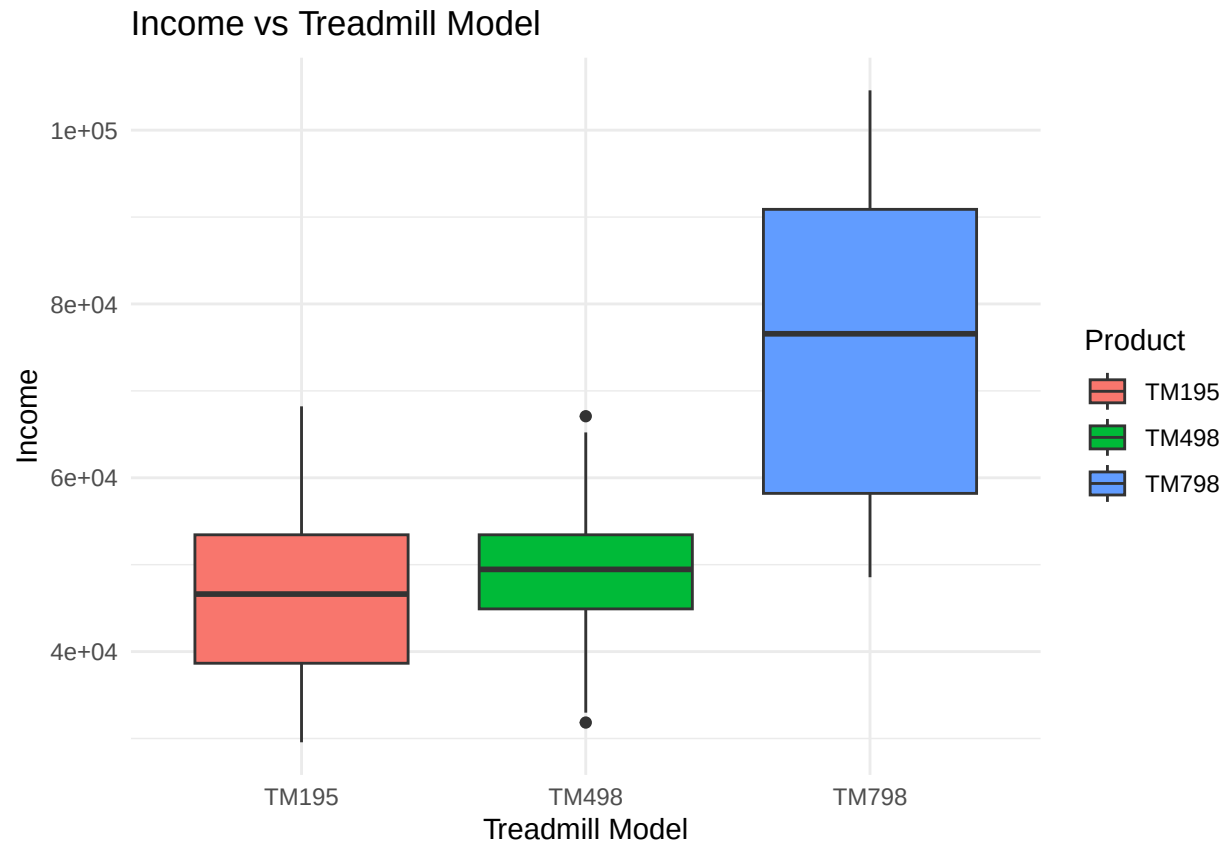
## 100	ProductTM195	Income	-0.396743770
## 101	ProductTM498	Income	-0.203871267
## 102	ProductTM798	Income	0.705367792
## 103	Age	Income	0.513413685
## 104	GenderMale	Income	0.202052581
## 105	Education	Income	0.625827346
## 106	MaritalStatusSingle	Income	-0.150292900
## 107	Usage	Income	0.519537232
## 108	Fitness	Income	0.535005317
## 109	Income	Income	1.000000000
## 110	Miles	Income	0.543473259
## 111	ProductTM195	Miles	-0.352914916
## 112	ProductTM498	Miles	-0.208649907
## 113	ProductTM798	Miles	0.658400761
## 114	Age	Miles	0.036617569
## 115	GenderMale	Miles	0.217868763
## 116	Education	Miles	0.307284282
## 117	MaritalStatusSingle	Miles	-0.025638980
## 118	Usage	Miles	0.759130477
## 119	Fitness	Miles	0.785701743
## 120	Income	Miles	0.543473259
## 121	Miles	Miles	1.000000000

```
ggplot(data = melted_cormat, aes(x = Var1, y = Var2, fill = value)) +
  geom_tile() +
  geom_text(aes(label = round(value, 2)), color = "black") +
  scale_fill_gradient2(low = "blue", high = "red", mid = "white", midpoint = 0, limit = c(-1, 1), space
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, vjust = 1, hjust = 1), legend.position = "none")
```



#Relationships bewtween income and models

```
ggplot(data, aes(x = Product, y = Income, fill = Product)) + geom_boxplot() + labs(title = "Income vs T
```

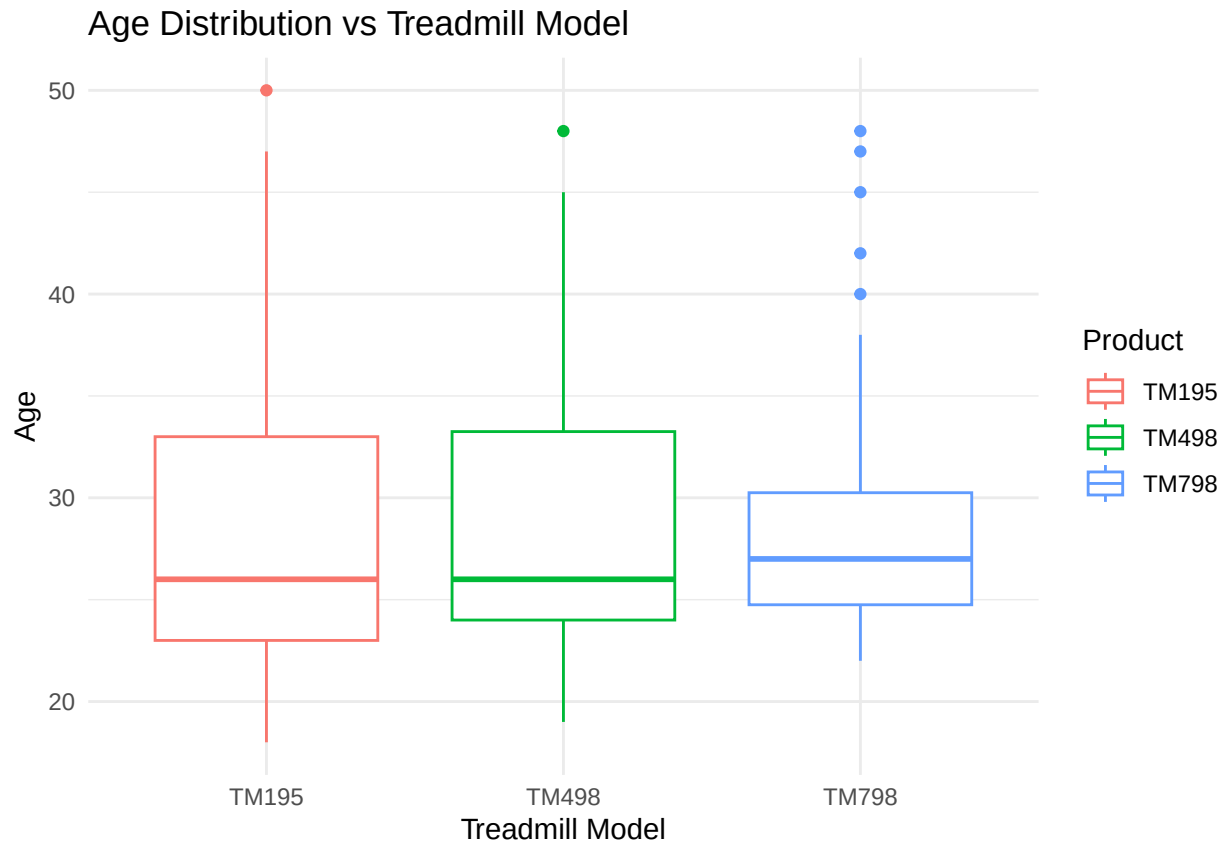


```
anova_income_model <- aov(Income ~ Product, data = data)
summary(anova_income_model)
```

```
##           Df    Sum Sq  Mean Sq F value Pr(>F)
## Product      2 2.449e+10 1.225e+10   89.26 <2e-16 ***
## Residuals  177 2.428e+10 1.372e+08
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

#Relationships between age and models

```
ggplot(data, aes(x = Product, y = Age)) +
  geom_boxplot(aes(color = Product)) +
  labs(title = "Age Distribution vs Treadmill Model", x = "Treadmill Model", y = "Age") +
  theme_minimal()
```

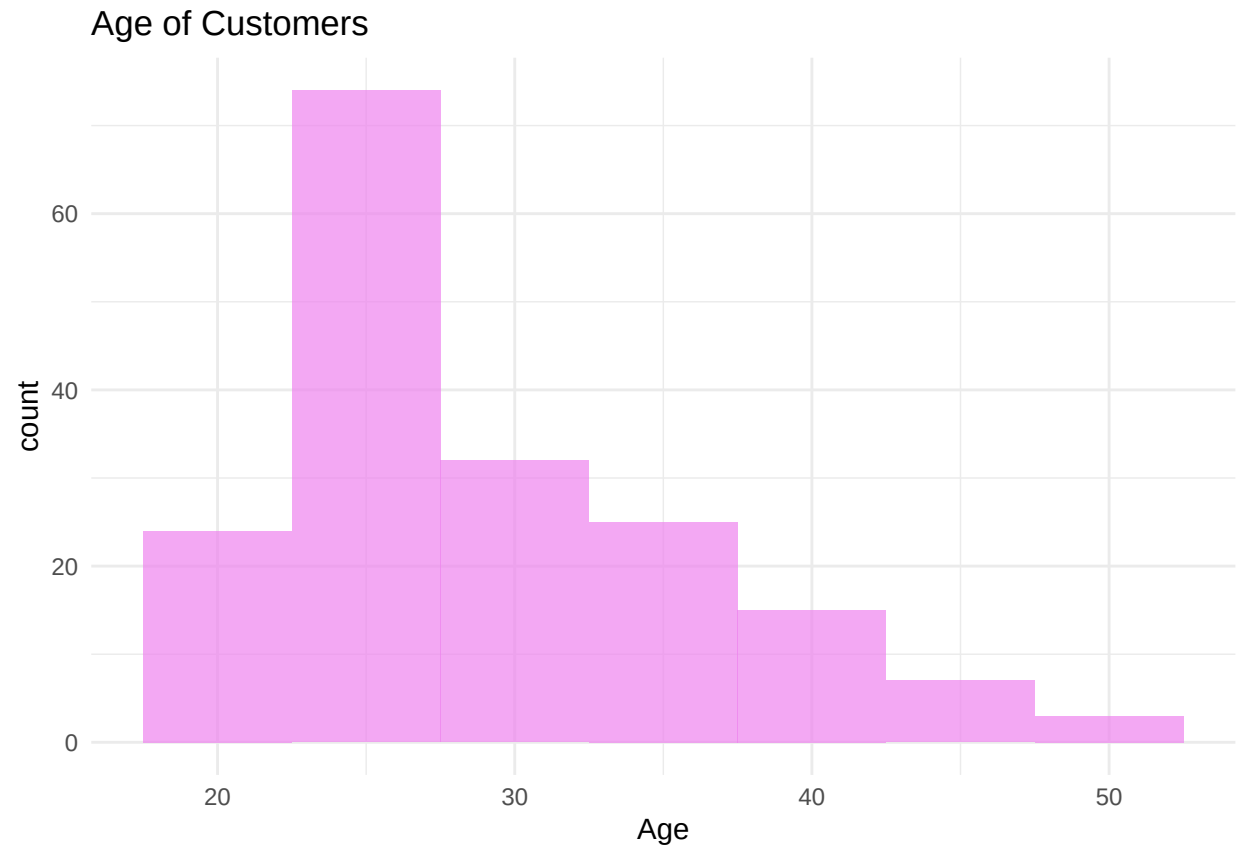


```
#Relationships between self-rated fitness and model
summary_by_model <- aov(Fitness ~ Product, data = data)
print(summary_by_model)
```

```
## Call:
##   aov(formula = Fitness ~ Product, data = data)
##
## Terms:
##               Product Residuals
## Sum of Squares  88.91528  75.66250
## Deg. of Freedom      2      177
##
## Residual standard error: 0.6538132
## Estimated effects may be unbalanced
```

```
#Relationships between education and model

ggplot(data, aes(x = Age)) +
  geom_histogram(binwidth = 5, fill = "violet", alpha = 0.7) +
  labs(title = "Age of Customers")+ theme_minimal()
```

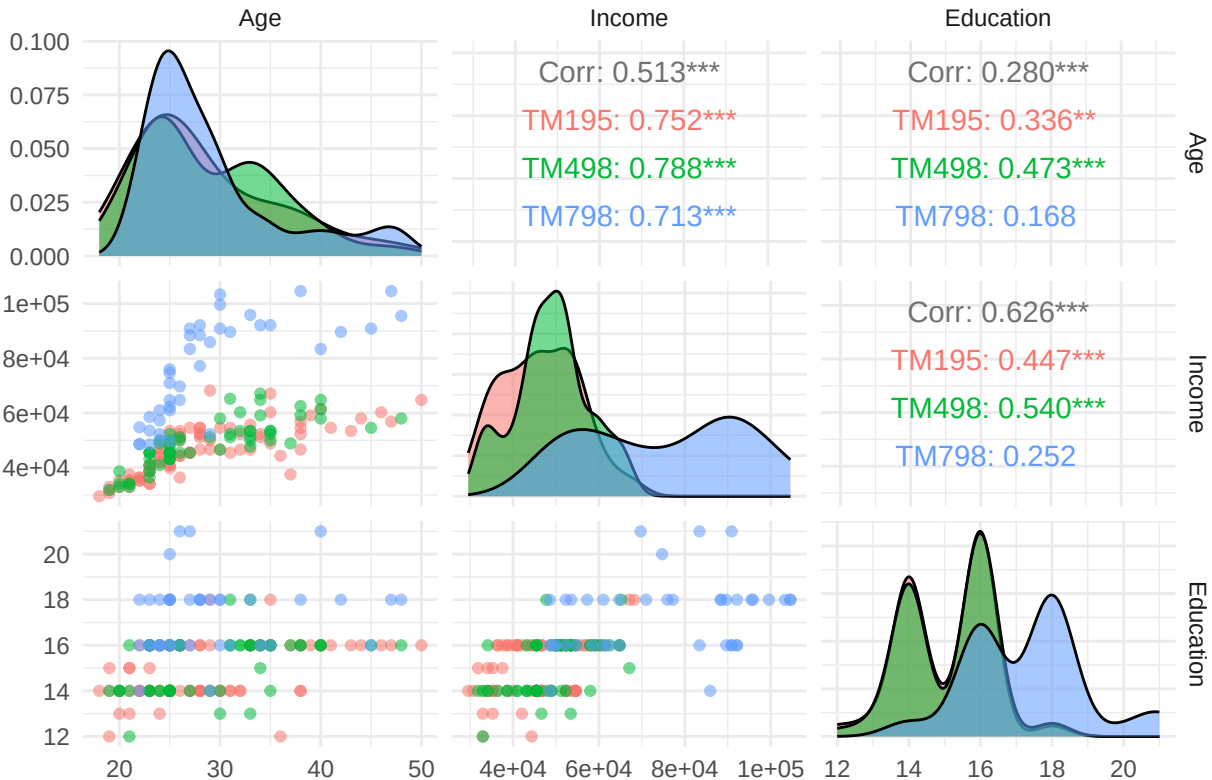


```
library(pairsD3)
library(GGally)
```

```
## Registered S3 method overwritten by 'GGally':
##   method from
##   +.gg    ggplot2
```

```
ggpairs(data, columns = c("Age", "Income", "Education"),
  aes(color = Product, alpha = 0.5),
  title = "Pair Plot of Age, Income, and Education by Treadmill Model")+ theme_minimal()
```

Pair Plot of Age, Income, and Education by Treadmill Model



```
# Chi square for Marital Status and Treadmill Model
chisq_test_marital <- chisq.test(table(data$Product, data$MaritalStatus))
print(chisq_test_marital)
```

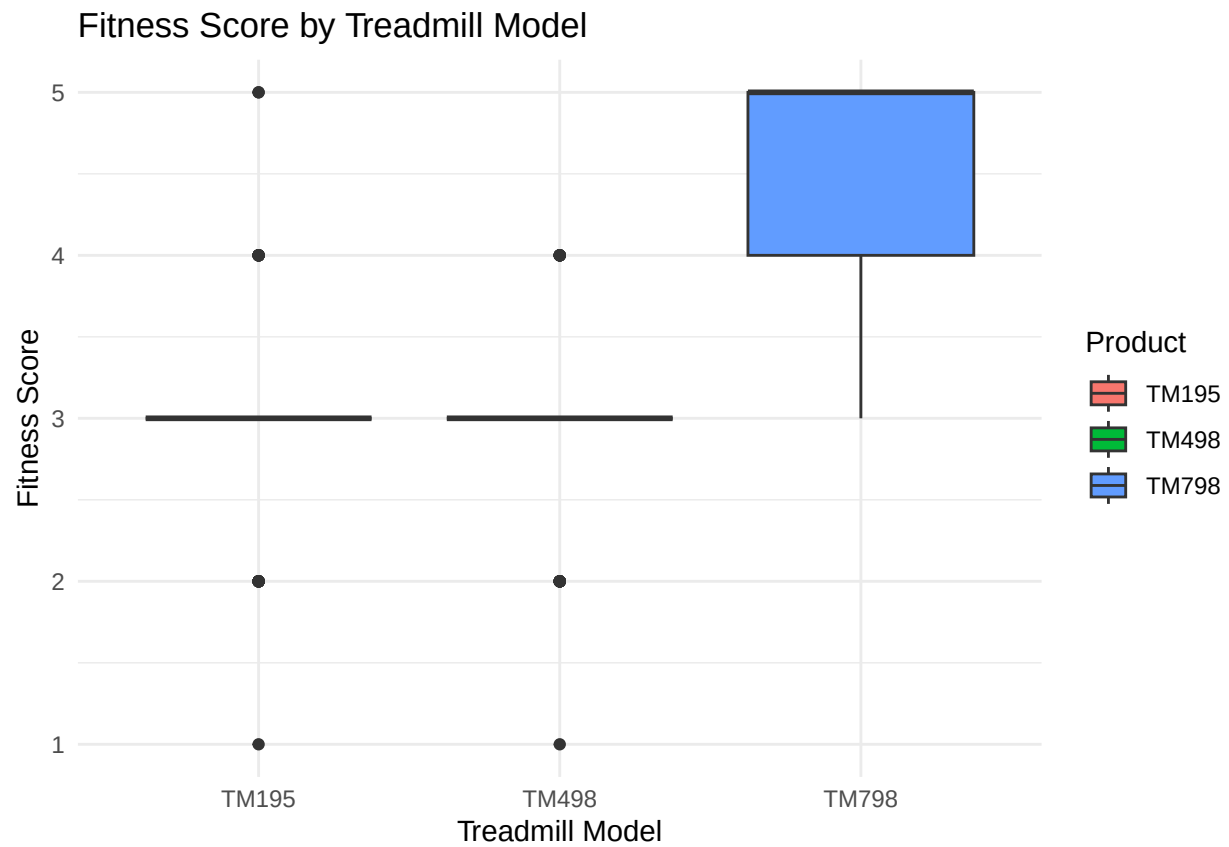
```
##
## Pearson's Chi-squared test
##
## data: table(data$Product, data$MaritalStatus)
## X-squared = 0.080655, df = 2, p-value = 0.9605
```

```
# Chi square for Gender and Treadmill Model
chisq_test_gender=chisq.test(table(data$Product, data$Gender))
print(chisq_test_gender)
```

```
##
## Pearson's Chi-squared test
##
## data: table(data$Product, data$Gender)
## X-squared = 12.924, df = 2, p-value = 0.001562
```

```
# boxplot of fitness vs product
```

```
ggplot(data, aes(x = Product, y = Fitness)) +
  geom_boxplot(aes(fill = Product)) +
  labs(title = "Fitness Score by Treadmill Model", x = "Treadmill Model", y = "Fitness Score")+ theme_m
```

#boxplot of weekly usage and product

```
ggplot(data, aes(x = Product, y = Usage)) +  
  geom_boxplot(aes(fill = Product)) +  
  labs(title = "Weekly Usage by Treadmill Model", x = "Treadmill Model", y = "Weekly Usage (Times)") + theme_minimal()
```

